

OLLITAL

TECHNICAL WHITE PAPER

# Solvent Recovery Technologies: A Comparative Guide

Engineering routes for closed-loop spray drying, vapor capture, inerting and controlled solvent handling.

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</solutions/organic-solvent-drying/>

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# Decision Brief

Solvent service changes the spray-drying design basis. The feed, vapor load, oxygen limit, condenser duty and recovery objective must be reviewed as one operating system. This guide compares common recovery routes and organizes the decisions required before equipment selection.

**The preferred route depends on solvent properties, expected vapor load, target recovery, utilities and the site's hazard assessment.**



## WHAT THE REVIEW SHOULD DECIDE

### 01 Define the solvent envelope

List composition, boiling behavior, flash point, water content and credible feed variation.

### 02 Match capture to the duty

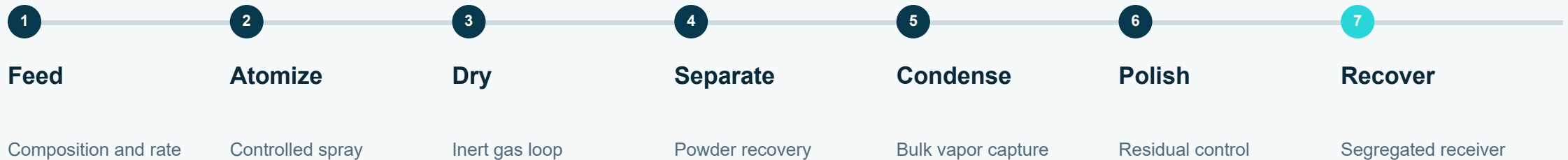
Use condensation for bulk recovery, then add deeper cooling or polishing only when the target requires it.

### 03 Engineer the whole loop

Dryer, condenser, receiver, inerting, controls and room interfaces must respond as one system.

# Closed-Loop Process Architecture

A coordinated loop controls oxygen, captures vapor and returns the system to a safe state when operating conditions move outside the approved window.



# Recovery Route Comparison

Each route serves a different vapor load and recovery target. The practical answer often combines bulk condensation with one finishing stage.



## Single-stage condensation

**Best fit: moderate vapor load**

Simple architecture with low equipment count. Low dew-point solvents may leave appreciable vapor in the return gas.



## Two-stage condensation

**Best fit: higher recovery targets**

Bulk capture followed by deeper cooling. Review utility demand, icing risk and condensate segregation.



## Vacuum-assisted recovery

**Best fit: lower temperature duty**

Can reduce boiling temperature and shift vapor behavior. Requires tightness, compatible pumps and oxygen control.



## Adsorption or polishing

**Best fit: low residual concentration**

Useful after bulk condensation. Media compatibility, breakthrough monitoring and regeneration drive lifecycle cost.



# Integrated Recovery Train



The equipment train should be defined around the operating envelope, not assembled as separate packages after the dryer has been selected.

## 01 Primary condenser

Removes the main solvent load at a practical coolant temperature.

## 02 Secondary cooling

Raises recovery when one stage cannot meet the selected residual target.

## 03 Inerting and controls

Maintain the approved oxygen window and coordinate permissives and shutdowns.

## 04 Receiver and segregation

Collect recovered solvent safely and avoid unintended mixing between campaigns.

# Engineering Variables



The comparison only becomes meaningful when the same material balance, temperature basis and operating limits are applied to every route.

## Solvent properties

Boiling range, vapor pressure, water miscibility and materials compatibility.

## Vapor load

Feed rate, solvent fraction, air or nitrogen flow and expected transient peaks.

## Recovery objective

Bulk reuse, emissions control, product protection or a combination of these goals.

## Utilities and site

Coolant levels, power, nitrogen, vent treatment and local hazardous-area requirements.

# Project Input Checklist

A complete solvent data package lets the engineering team compare realistic routes instead of relying on generic recovery percentages.



## Feed basis

- Solvent composition
- Solids and viscosity
- Water content
- Expected variation

## Process target

- Evaporation rate
- Temperature limit
- Residual solvent
- Powder yield

## Safety basis

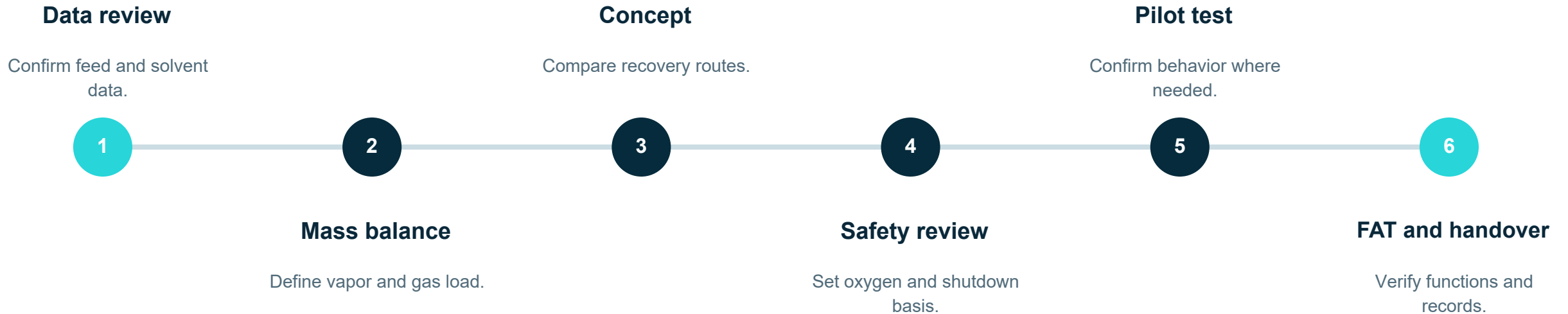
- Flash point and LEL
- Oxygen limit
- Area classification
- Exposure limits

## Site interface

- Coolant temperatures
- Nitrogen supply
- Vent treatment
- Receiver handling

# Project Validation Path

A staged review develops the evidence needed for safe operation and a defensible recovery estimate.



**State the operating basis beside every predicted recovery value. Solvent composition, coolant temperature and gas flow can materially change the result.**



# Safety and Review Framework

The final design should align with the site's process safety method and the applicable market requirements. These references help structure the review without replacing project-specific engineering.

- |           |                                 |                                                                                                 |
|-----------|---------------------------------|-------------------------------------------------------------------------------------------------|
| <b>01</b> | <b>ATEX equipment direction</b> | Review equipment category and protection concepts where explosive atmospheres may occur.        |
| <b>02</b> | <b>Hazard assessment</b>        | Document credible releases, ignition sources, oxygen conditions and safeguards.                 |
| <b>03</b> | <b>Functional response</b>      | Define alarms, permissives, trips and the safe state for utilities or instrumentation failures. |
| <b>04</b> | <b>Lifecycle documentation</b>  | Maintain material records, certificates, inspection evidence, FAT results and change control.   |



# Turn the solvent data into a recovery concept

Share the feed composition, target throughput, recovery objective, temperature limits and site utilities. OLLITAL can review the most appropriate laboratory, pilot or production route.

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## MATERIAL EVALUATION

<https://ollitaldryer.com/material-evaluation/#evaluation-form>

## CONTACT

meiko@ollitaldryer.com +86 159 6082 1529

No. 39, Xinchang Road, Haicang District, Xiamen, Fujian, China

**OLLITAL**

A large, complex industrial solvent recovery system made of stainless steel. It features several large vertical cylindrical tanks, a network of pipes with valves, and a horizontal cylindrical vessel at the bottom. A control panel with a digital display is visible on the right side of the unit. The background is a dark, industrial setting with blue lighting.